

Documenting Changes in 01_download/README.Rmd

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0.1 Purpose

Provide a human-readable summary of all download scripts in R/01_download/, including their purpose, input/output expectations, status, and key changes over time. This supports reproducibility and onboarding of new contributors.

0.2 Overview

This section documents the current structure and status of all scripts responsible for downloading raw spatial and tabular data inputs. Each script is version-controlled and tied to a specific covariate or source dataset (See Table 1).

Table 1: Script Index

File Name	Description	Output Folder	Notes / Status
01a_download_ecoregions.R	Download Level I-IV US ecoregion shapefiles	data/raw/	CRS standardized (EPSG:5070)
01b_download_USGS_gage_data.R	Download peak flow gage locations	data/raw/	Reorganized metadata
01c_download_usgs_gage_metadata.R	Query detailed site metadata	data/raw/	Outputs raw data-dictionary
01d_download_usgs_peakflow_data.R	Download annual peak flow values	data/raw/	Handles chunking & retries
01e_filter_usgs_peakflow_data.R	Filter data, calculate skew, tag PILFs	data/processed/	Replaces legacy script
01f_download_nhdplus_v21.R	Download NHDPlusV2.1 flowlines and catchments	data/processed/	CRS standardized (EPSG:5070)

0.3 Summary of Key Changes

0.3.1 Standardized Folder Paths

- All output paths now use `here::here("data", "raw", ...)` or `data/processed/`
- Ensures reproducibility across systems and OSes

0.3.2 Consistent CRS

- All spatial outputs reprojected to EPSG:5070 (CONUS Albers Equal Area)

0.3.3 Modular Design

- Each script targets a single source or covariate
- Utility functions moved to `R/utils/`

0.3.4 Logging & QA

- Retry attempts, download failures, and chunking behavior logged to `.csv` where applicable
 - Scripts now include `{cli}` alerts for status tracking
 - Each script is cross-linked to relevant milestone reports
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